

Econ 131
Spring 2026

Midterm *Solutions*

March 2026

Student Name:

Student ID:

Exam Instructions

- **This exam has three questions. Each question is worth 50 points (150 points total). The total exam time is 75 minutes.**
- **Write answers using pens (we recommend black or blue ink), and write only in the space provided for each answer so your solutions scan properly.**
- **Do not write your solutions on pages that say “Do not write on this page”. Answers written on these pages will not be graded.**
- **Show your work.** Credit will only be awarded on the basis of what is written on the exam.
- **Affirm the academic honesty pledge below.** For those writing on a non-printed copy, please write “Academic Honesty Pledge as on exam” and sign your name. **If you do not affirm this pledge, your exam will be marked invalid.**

0. ACADEMIC HONESTY PLEDGE

I confirm that I have abided by all academic honesty rules for UC Berkeley and Economics 131. I confirm that I did not see this exam before my official exam start time. I confirm that I have not shared and will not share this exam with anyone else.

Signature: _____

1. True/False/Uncertain Questions (50 points total)

- (a) Redistribution is one reason for government intervention in the economy.

True. Redistribution is a central rationale for policy in public finance. If marginal utility of consumption declines with income, transferring one dollar from a high-income to a low-income household can raise utilitarian social welfare. This equity motive exists even when markets are otherwise competitive.

- (b) In a simple observational correlation, a large sample is enough to make the estimate causal.

False. A larger sample mainly reduces variance and narrows confidence intervals. It does not fix core identification threats such as omitted variables, selection, or reverse causality. So a very precise estimate can still be precisely wrong about the causal effect.

- (c) If the GDP growth rate is higher than the interest rate on government debt, then the government can run a primary deficit while keeping debt/GDP stable.

True. When $g > r$, the existing debt stock grows more slowly than the tax base (GDP), which mechanically lowers debt/GDP over time absent new deficits. That creates fiscal space for a sustained positive primary deficit while still stabilizing the ratio. The exact sustainable deficit depends on the current debt ratio and the $g - r$ gap.

- (d) If taxes and transfers have absolutely no behavioral effects, a utilitarian government sets taxes and transfers so that everyone consumes the same amount.

True. With zero behavioral responses, there is no efficiency cost to redistribution. A utilitarian planner with concave utility maximizes welfare by equalizing marginal utilities across people; with identical utility curvature, that implies equalizing consumption levels. In that knife-edge case, full equalization is first-best.

- (e) An Earned Income Tax Credit is desirable when the intensive margin response is larger than the extensive margin response.

False. The strongest efficiency argument for an EITC is when it raises labor-force participation (extensive margin), because moving nonworkers into work can expand earnings and the tax base. If responses are mainly intensive (hours/earnings among workers), phase-in and phase-out regions create stronger distortionary responses. So a larger intensive response relative to extensive generally weakens, not strengthens, the standard EITC case.

2. Incidence of Commodity Taxation

Consider the following stylized market for televisions in Berkeley. Let market demand and supply be summarized as

$$Q^D = 912 - \frac{P}{5}, \quad Q^S = \frac{P}{10},$$

where P is the dollar price per TV and Q is measured in thousands of TVs.

- (a) Solve for the competitive equilibrium before any tax. Report equilibrium price and quantity. (10 points)

Pre-tax equilibrium:

$$Q^D = Q^S \Rightarrow 912 - \frac{P}{5} = \frac{P}{10} \Rightarrow 912 = \frac{3P}{10} \Rightarrow P_0 = 3040, \quad Q_0 = 304.$$

Quantity is in thousands of TVs.

- (b) Now impose a per-unit tax of $t = \$60$ on buyers. Solve for the post-tax equilibrium and report the consumer price, producer price, and quantity. (10 points)

Tax equilibrium: Let q be consumer price and p be producer price, so $q = p + 60$.

$$Q^D = 912 - \frac{q}{5} = 912 - \frac{p + 60}{5}, \quad Q^S = \frac{p}{10}.$$
$$912 - \frac{p + 60}{5} = \frac{p}{10} \Rightarrow 900 = \frac{3p}{10} \Rightarrow p = 3000, \quad q = 3060, \quad Q = \frac{p}{10} = 300.$$

- (c) Compute the deadweight loss from the tax and illustrate it on a supply-demand graph. (10 points)

Deadweight loss:

$$DWL = \frac{1}{2} \cdot t \cdot (Q_0 - Q) = \frac{1}{2} \cdot 60 \cdot (304 - 300) = 120.$$

Since Q is in thousands, this is in thousands of dollars.

- (d) Compute the tax incidence on consumers and producers, and explain intuitively which side bears more of the burden and why. (10 points)

Incidence:

$$\text{Consumer burden per TV} = q - P_0 = 3060 - 3040 = 20,$$

$$\text{Producer burden per TV} = P_0 - p = 3040 - 3000 = 40.$$

Consumers bear 1/3 and producers bear 2/3 of the tax. Producers bear more because supply is less elastic than demand.

- (e) Suppose consumers are fully inattentive to the tax and keep using the original demand equation as if no tax exists. What does this imply for deadweight loss and incidence? You do not need to do any math; full credit will be given for clear intuition. (10 points)

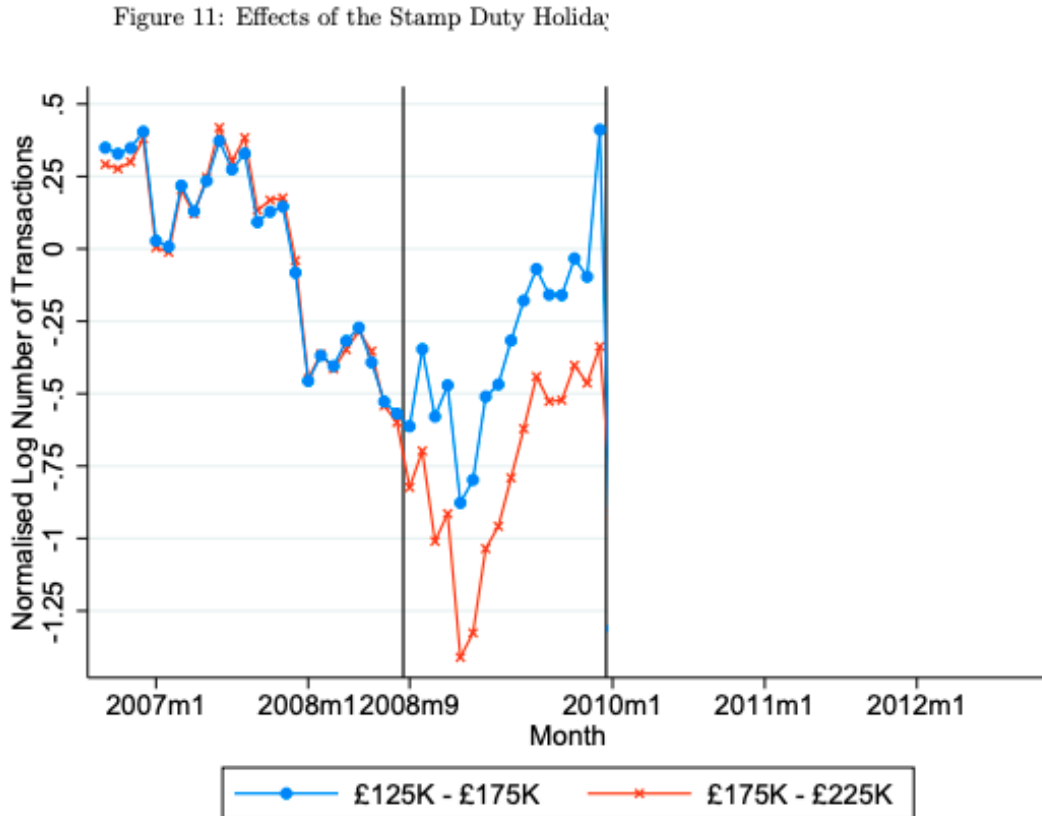
If consumers are fully inattentive and keep using the original demand equation as if no tax exists, quantity does not fall from its pre-tax level, so deadweight loss is zero. The full tax burden shifts to consumers through the tax-inclusive price.

3. Empirical Application: Best and Kleven

In the the United Kingdom, the government taxes home sales. In response to the Great Recession, the UK government in September 2008 eliminated the tax on homes in the sale price range between £125,000 and £175,000. This temporary tax break lasted for 16 months, from September 2008 through December 2009. The tax rate on properties in the sale price range between £175,000 and £225,000 remained unchanged.

- (a) Suppose you find that in the 16-month period, more homes were sold in the £125,000-£175,000 price range than in the £175,000-£225,000 price range. Is that good evidence that the tax break caused an increase in home sales in the £125,000-£175,000 price range, why or why not? (12.5 points)

No, that alone is not good causal evidence. A level comparison in one period does not net out pre-existing differences in market size, composition, or secular trends between the two price ranges. The treatment range could always have had more sales even without the reform. Causal interpretation requires comparing changes over time relative to a credible control group, not raw levels.



- (b) Suppose you draw the figure above. The vertical black line indicates September 2008. The circles show the number of home sales in the £125,000-£175,000 price range, expressed as the percent change from early 2007. The stars show the number of home sales in the £175,000-£225,000 price

range, also expressed as the percent change from early 2007. Is this good evidence that the tax break caused an increase in home sales in the £125,000-£175,000 price range, why or why not? (12.5 points)

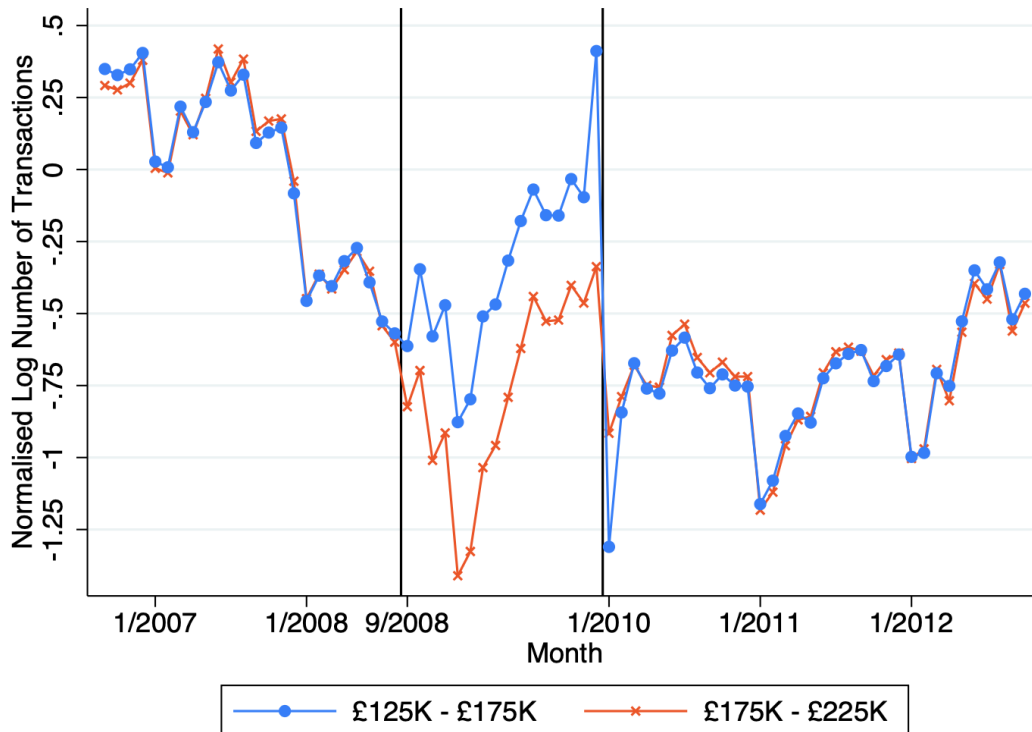
Yes, this is substantially more persuasive. The pre-reform paths look very similar, which supports the parallel-trends assumption needed for a difference-in-differences design. Right after September 2008, a clear gap opens with the treated range rising relative to the control range, matching the timing of the policy. That timing pattern is what we would expect if the tax cut increased transactions in the treated bracket.

- (c) Looking at the graph, what do you think a difference-in-differences table would say is the difference-in-differences estimate of the causal effect of the tax break on home sales in the £125,000-£175,000 price range? (Concretely, you can imagine that you have the average value of the circles before the tax break, the average value of the stars before the tax break, the average value of the circles after the tax break, and the average value of the stars after the tax break. And that you use those four numbers to compute the difference-in-differences estimate.) (12.5 points)

The DiD estimate should be positive and economically meaningful. Conceptually:

$$\hat{\beta}^{DID} = (\bar{Y}_{T,after} - \bar{Y}_{T,before}) - (\bar{Y}_{C,after} - \bar{Y}_{C,before}).$$

From the graph, the pre-reform gap appears close to zero, while during the reform the treated series is noticeably above the control series, so the estimate should be approximately the average vertical treated-control gap during the policy window.



- (d) Now suppose you extend the figure to include data points after December 2009, as in this second figure. Does this new figure provide additional comforting evidence that the tax break caused an increase in home sales in the £125,000-£175,000 price range, why or why not? *(12.5 points)*

Yes, this adds important dynamic evidence. If the treated-control gap narrows or disappears once the temporary tax break ends, that “on-when-treated, off-when-removed” pattern strongly supports a causal interpretation and weakens alternative stories based on permanent unobserved differences. In other words, the event-time pattern shows strong pre-trend similarity, a reform-period divergence, and post-reform convergence, which is exactly the signature expected from a temporary policy effect.

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